

Signal and System Design (IGS2137)

Jump Together, Fly Farther!



인하대학교 국제학부

Week 10 Lab

**ISE Department
Prof. Mehdi Pirahandeh**

Midterm Review Exercise:

Goals :

1. Repeat the same operations (shifting, scaling, reversing)
2. Write a report explaining each step of the implementation
3. Understand and implement algorithms with MATLAB

Midterm Review Exercise:

Task:

- Make a program to perform x3 operations as shown below:
- The user should input the value as:
 - a) Type of operation(shifting, scaling, reversing)
 - b) Type of Signal (Discrete or Continuous)
 - c) Amount of shift or value based on the type of operation user selected

Midterm Review Exercise:

Assumptions:

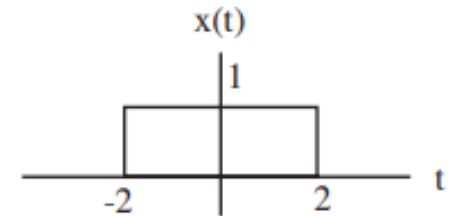
- Amount of shift/scaling/reversing can be + or -
- Exception should be thrown in case the user input an unrealistic value

Challenge:

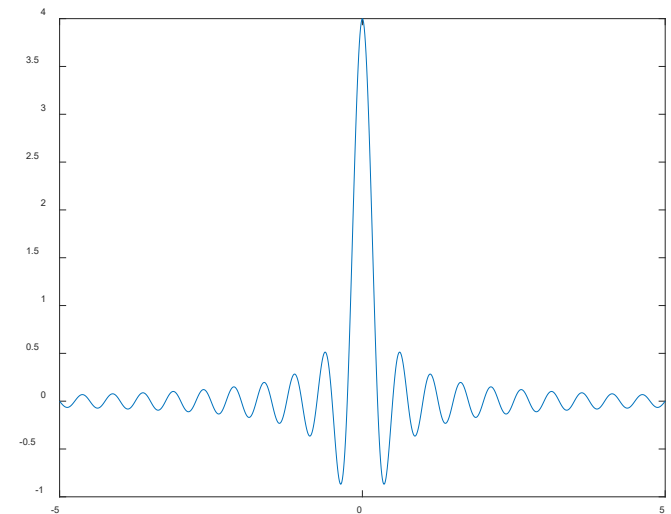
- You need to select 4 different signals instead of the 3 used in lecture material
- One must of sinusoidal wave (sine or cosine)
- One must be Saw tooth or Square signal
- Other two can be any of your choices
- Implementing using App Designer

Frequency Domain Analysis

Exercise 1: Write a script that will the following step and plot the given function. The aperiodic pulse is shown below



has a Fourier transform: $X(jf) = 4\text{sinc}(4\pi f)$



Frequency Domain Analysis

```
f=-5:.01:5;  
X=4*sinc(4*f);  
figure, plot(f,X)  
figure,plot(f,abs(X))  
figure,plot(f,angle(X))
```

- In this case, the Fourier transform is a purely real function. Thus, we can plot it as shown above.
- In general, fourier transforms are complex functions and we need to plot the amplitude and phase spectrum separately

Frequency Domain Analysis

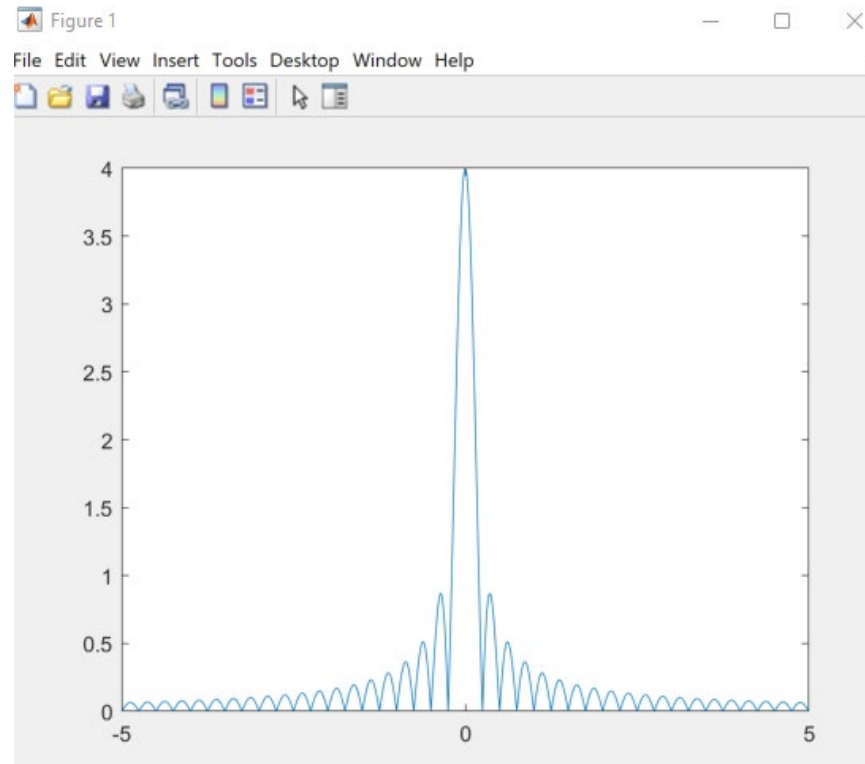
```
f=-5:.01:5;  
X=4*sinc(4*f);  
figure, plot(f,X)  
figure, plot(f,abs(X))  
figure, plot(f,angle(X))
```

$\theta = \text{angle}(z)$ returns the phase angle in the interval $[-\pi, \pi]$ for each element of a complex array z . The angles in θ are such that $z = \text{abs}(z) \cdot \exp(i \cdot \theta)$.

- In this case, the Fourier transform is a purely real function. Thus, we can plot it as shown above.
- In general, fourier transforms are complex functions and we need to plot the amplitude and phase spectrum separately

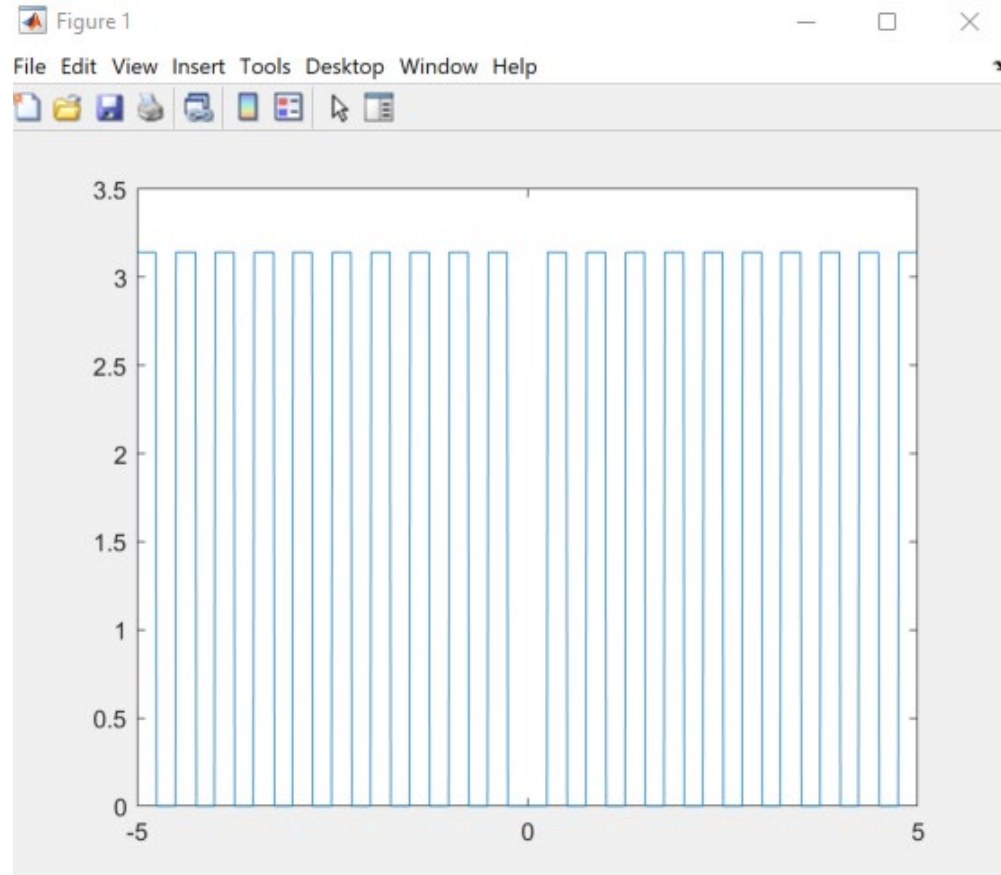
Frequency Domain Analysis

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Frequency Domain Analysis

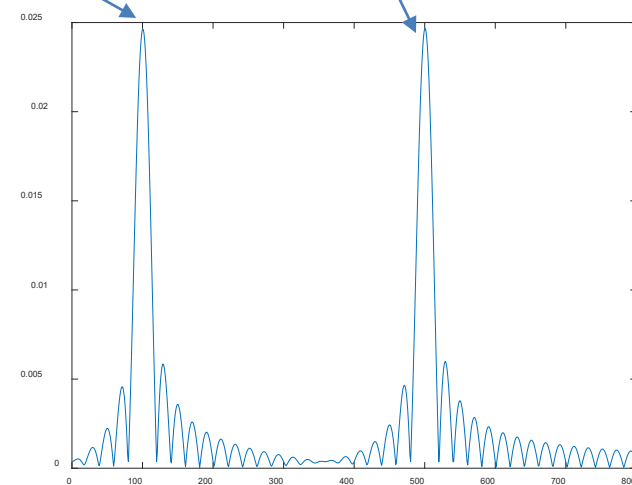
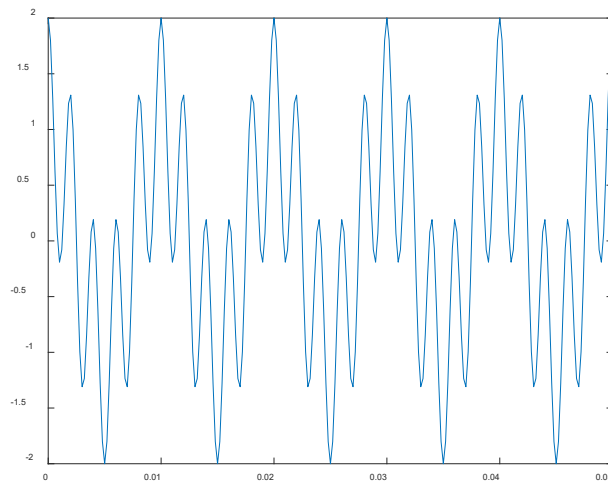
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f=-5:.01:5;  
X=4*sinc(4*f);  
figure, plot(f,X)  
figure,plot(f,abs(X))  
figure,plot(f,angle(X))
```



Frequency Domain Analysis

Exercise 2: Write a script that will the following step and plots the given function. Multiple Frequency Signals is shown below

$$x(t) = \cos(2\pi 100t) + \cos(2\pi 500t)$$



Frequency Domain Analysis

%%% finding integral in Matlab

f=0;

t=-2:.01:2;

trapz(t, exp(-j*2*pi*f*t))

%% trapz is to find the integral ,

$$X(jf) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

Frequency Domain Analysis

%%% for complete integral

t=-2:.01:2;

k=0;

for f=-5:.01:5

k=k+1;

X(k)=trapz(t,exp(-j*2*pi*f*t));

end

f=-5:.01:5;

plot(f,X)

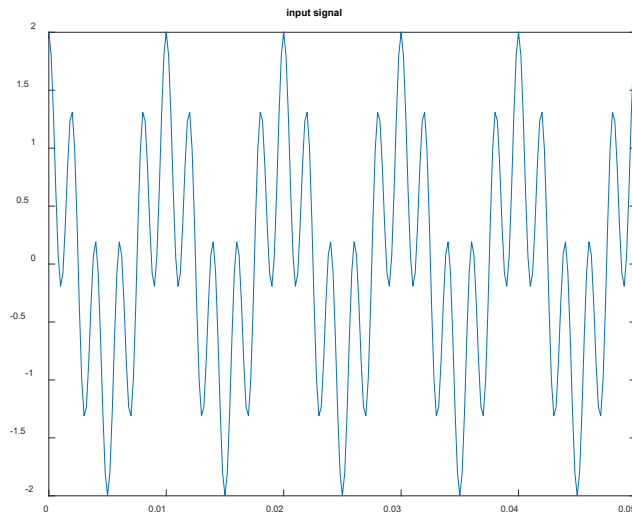
$$X(jf) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$X(jf) = \int_{-2}^2 1 e^{-j2\pi ft} dt$$

Frequency Domain Analysis

Write a script that will the following step and plots the given function. Multiple Frequency Signals is shown below

$$x(t) = \cos(2\pi 100t) + \cos(2\pi 500t)$$

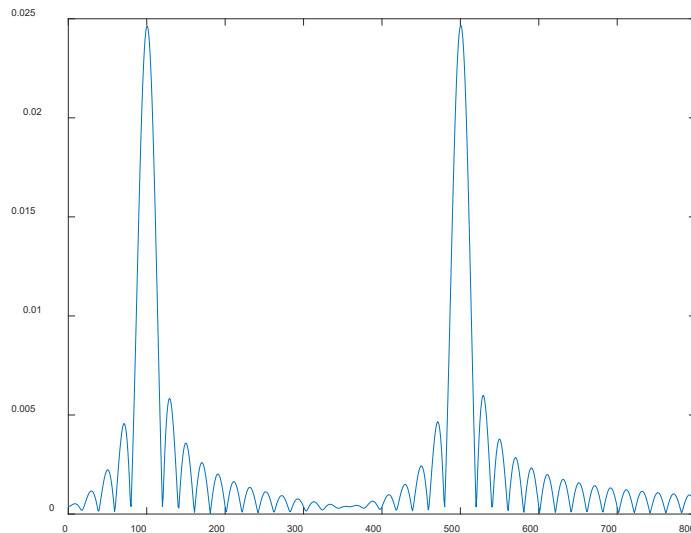


```
N=250;  
ts=.0002;  
t=[0:N-1]*ts;  
x=cos(2*pi*100*t)+cos(2*pi*500*t);  
figure, plot(t,x), title ('input signal')
```

Frequency Domain Analysis

Write a script that will the following step and plots the given function. Multiple Frequency Signals is shown below

$$x(t) = \cos(2\pi 100t) + \cos(2\pi 500t)$$



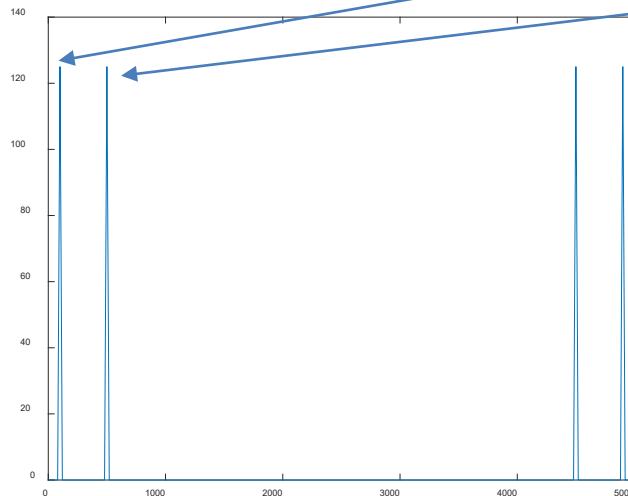
```
% We can find approximate the Fourier transform integral for 0 ≤ f ≤ 800 Hz using:
%clear all;
%clc
k = 0;
for f = 0:1:800
    k=k+1;
    %fourier transform formula
    X(k) = trapz(t,x.*exp(-j*2*pi*f*t));
end

f=0:800;
figure
plot(f,abs(X));
title('Fourier transform')
```

Frequency Domain Analysis

Write a script that will the following step and plots the given function. Multiple Frequency Signals is shown below

$$x(t) = \cos(2\pi 100t) + \cos(2\pi 500t)$$



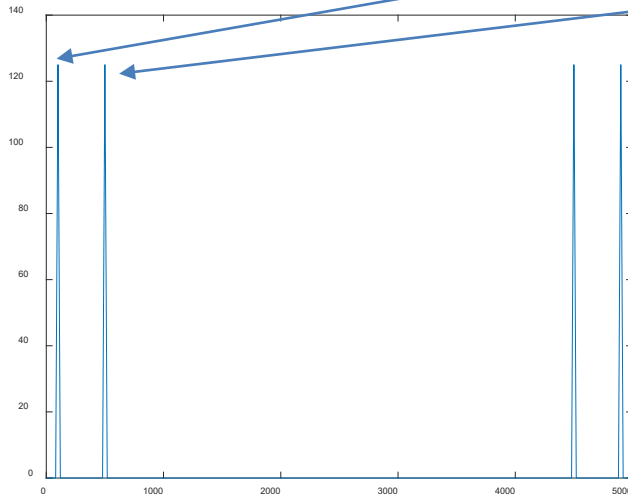
- Note that the spectrum shows four components.
- Two are at the expected frequencies of 100 and 500 Hz.
- The other two are at 4500 and 4900 Hz, frequencies that do not appear in the signal.
- This is due to the periodic nature of the DFT.
- Only frequencies up to $0.5/T_s$ correspond to the actual frequencies in the Fourier transform.

DFT → FFT are the same
FFT is an optimized method to implement DFT

Frequency Domain Analysis

Write a script that will the following step and plots the given function. Multiple Frequency Signals is shown below

$$x(t) = \cos(2\pi 100t) + \cos(2\pi 500t)$$

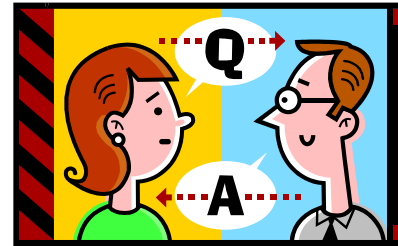


DFT → FFT are same
FFT is optimized method to implement DFT

```
% DFT
N=250;
ts=.0002;
deltaf=1/(N*ts);
t=[0:N-1]*ts;
x=cos(2*pi*100*t)+cos(2*pi*500*t);
Xf=fft(x);
f=[0:N-1]*deltaf;
figure,plot(f,abs(Xf))
```

Brief break (if on schedule)

Q&A?



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